

Student ID:

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Quiz no. 1

Winter Semester 2024-2025

DURATION: 30 MIN

FULL MARKS: 20

CSE 4107: Structured Programming I

Figures in the right margin indicate marks.

1. List out and explain the errors in the following code:

```
#include<stdio.h>;

void main(void){
    int a;
    float b;
    double c;

    print("Enter the values of a, b and c: ");
    scanf("%d %f %lf", a, b, c);
    printf("a = %d/n b = %f/n c = %lf/n", a, b, c);

    return 0;
};
```

7
(CO2)
(PO1)

2. Find the output of the following program:

```
#include<stdio.h>

void main (){
    int i=0, j=1, k=2;
    k = ++i + --k * -++j - 2;
    printf("%d %d %d", i, j, k--);

    int a=2, b=3, c=3, d=4, e;
    e = d += b *= c = a += -1;
    printf("%d %d %d %d %d", a, b, c, d, e);
}
```

4
(CO1)
(PO1)

3. Write a C program that takes the following three floating point numbers as inputs from the user:

- The radius of the cylinder
- The height of the cylinder
- Price of paint per unit area

6
(CO3)
(PO1, PO2)

Your program should output the price of painting the whole outer area of the cylinder (including the bottom base and top lid) showing the result upto 4 decimal places. The surface area of a cylinder is calculated using the following formula: , and the area of the top lid and bottom base are both calculated by the same formula: . The total cost should be calculated as:

4. Show the output for the following statement:

```
printf("%-3.2d %5.2f %2.3f", 2, 3.141, 12.02);
```

3
(CO1)
(PO1)